

|  |                                                                       | Cluster                                                                          |             |             |             |
|--|-----------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|-------------|-------------|
|  |                                                                       | Other                                                                            |             | Positive    |             |
|  |                                                                       | Direction                                                                        |             |             |             |
|  |                                                                       | down                                                                             |             | up          |             |
|  |                                                                       | P Value                                                                          |             |             |             |
|  |                                                                       | <div><div></div><div>&gt;= 0.05</div><div></div><div>&lt; 0.0001</div></div>     |             |             |             |
|  |                                                                       | RRA                                                                              |             |             |             |
|  |                                                                       | <div><div></div><div>no significant</div><div></div><div>significant</div></div> |             |             |             |
|  | HP-UTERUS-DIDELPHYS                                                   | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-SPLENIC-RUPTURE                                                    | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-RENAL-MAGNESIUM-WASTING                                            | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-PERINEAL-HYPOSPADIAS                                               | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-OSSEOUS-FINGER-SYNDACTYLY                                          | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-INCREASED-SERUM-TESTOSTERONE-LEVEL                                 | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-INCREASED-CIRCULATING-RENIN-CONCENTRATION                          | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-IMPAIRED-MYOCARDIAL-CONTRACTILITY                                  | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-HYPOCHLOREMIA                                                      | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-FETAL-POLYURIA                                                     | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-DISTAL-RENAL-TUBULAR-ACIDOSIS                                      | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-BLIND-VAGINA                                                       | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-APLASIA-HYPOPLASIA-OF-THE-FALLOPIAN-TUBE                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-ANDROGEN-INSUFFICIENCY                                             | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-ABSENT-PUBIC-HAIR                                                  | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-ABNORMALITY-OF-THE-EPIPHYSES-OF-THE-PROXIMAL-PHALANGES-OF-THE-HAND | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-ABNORMAL-GLOMERULAR-BASEMENT-MEMBRANE-MORPHOLOGY                   | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | HP-3-4-FINGER-OSSEUS-SYNDACTYLY                                       | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOMF-RETINOL-BINDING                                                  | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOMF-PROLINE-SODIUM-SYMPORTER-ACTIVITY                                | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOMF-OPSIN-BINDING                                                    | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOMF-AMMONIUM-CHANNEL-ACTIVITY                                        | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOCC-SPHERICAL-HIGH-DENSITY-LIPOPROTEIN-PARTICLE                      | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOCC-NEUROFIBRILLARY-TANGLE                                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-URETERIC-BUD-FORMATION                                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-TYPE-IV-HYPERSENSITIVITY                                         | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-SPONTANEOUS-SYNAPTIC-TRANSMISSION                                | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-RETINAL-PIGMENT-EPITHELIUM-DEVELOPMENT                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-REGULATION-OF-SKELETAL-MUSCLE-TISSUE-REGENERATION                | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-REGULATION-OF-THYROID-HORMONE-GENERATION                         | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-REGULATION-OF-SPONTANEOUS-SYNAPTIC-TRANSMISSION                  | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-REGULATION-OF-MESONEPHROS-DEVELOPMENT                            | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-POSITIVE-REGULATION-OF-SKELETAL-MUSCLE-TISSUE-REGENERATION       | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-PARACELLULAR-TRANSPORT                                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-OTIC-VESICLE-DEVELOPMENT                                         | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-NOREPINEPHRINE-BIOSYNTHETIC-PROCESS                              | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-NEPHRIC-DUCT-FORMATION                                           | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-MESONEPHRIC-TUBULE-FORMATION                                     | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-MAST-CELL-DIFFERENTIATION                                        | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-MAMMARY-GLAND-BRANCHING-INVOLVED-IN-PREGNANCY                    | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-MAINTENANCE-OF-GASTROINTESTINAL-EPITHELIUM                       | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-LOBAR-BRONCHUS-EPITHELIUM-DEVELOPMENT                            | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-LOBAR-BRONCHUS-DEVELOPMENT                                       | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-INTERCELLULAR-TRANSPORT                                          | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-ERBB4-SIGNALING-PATHWAY                                          | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-ERBB2-ERBB4-SIGNALING-PATHWAY                                    | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-DETOXIFICATION-OF-NITROGEN-COMPOUND                              | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-CENTRAL-NERVOUS-SYSTEM-MYELIN-MAINTENANCE                        | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-CELLULAR-RESPONSE-TO-POTASSIUM-ION                               | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |
|  | GOBP-AMMONIUM-TRANSMEMBRANE-TRANSPORT                                 | <div></div>                                                                      | <div></div> | <div></div> | <div></div> |